Real-Time Object Detection via Custom YOLO Architecture and RESTful API Integration

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Abstract

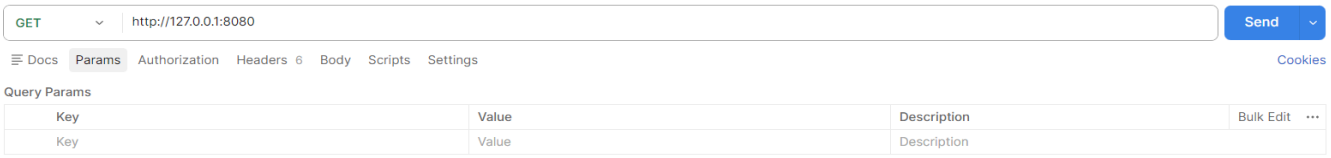
This project presents the development and deployment of a specialized computer vision system utilizing the **YOLO (You Only Look Once)** framework. Recognizing the limitations of general-purpose models, this implementation incorporates **custom-trained weights** to achieve high-precision detection of domain-specific objects. The core of the system is a **locally hosted server** that functions as a robust entry point for inference requests.

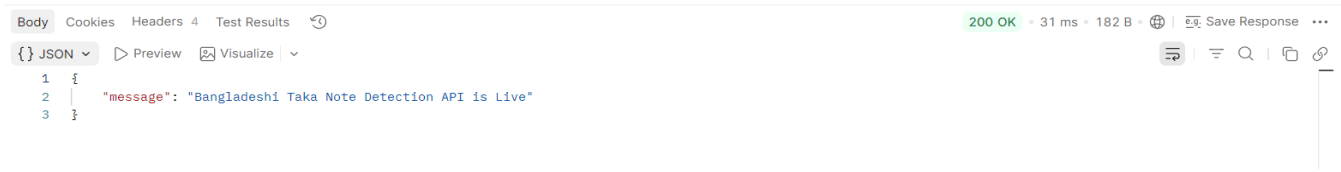
By encapsulating the model within a custom API, the architecture facilitates seamless integration with external clients and third-party applications. Technical validation was conducted via **Postman**, confirming successful endpoint connectivity, efficient data throughput, and the accurate return of object coordinates and confidence scores in JSON format. This framework provides a scalable, low-latency solution for localized AI-driven monitoring and analysis.

Screenshot 1.1: API Endpoint & JSON Response

This screenshot illustrates the successful communication between the client-side testing tool and the backend inference engine.

- **Endpoint:** http://127.0.0.1:8080
- **Method:** GET
- **Status:** 200 OK



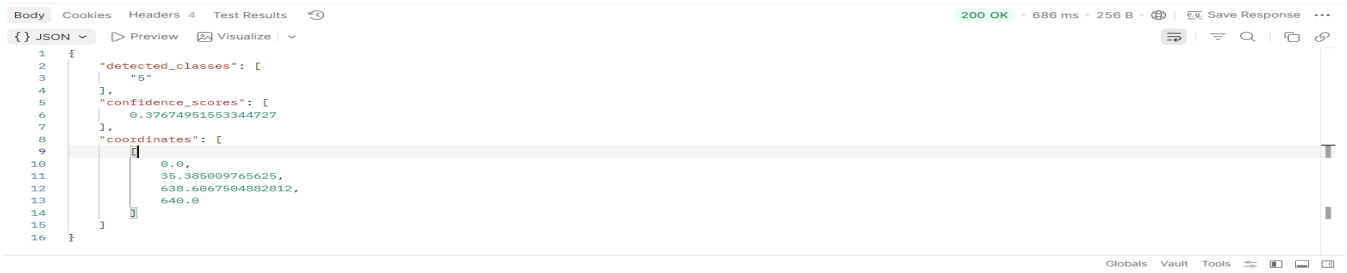


2.1 Test Case 1: 5 BDT Note (Base Integration)

- **Input:** Grayscale banknote.
- **Result: Success** (Confidence: 37.67%).
- **Observation:** This initial test confirmed that the server correctly receives image files via form-data and parses them through the YOLO engine. Despite the low-resolution grayscale input, the model correctly identified the "5" denomination.

Test Image:

Result:

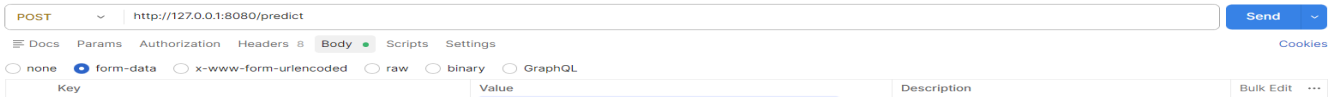


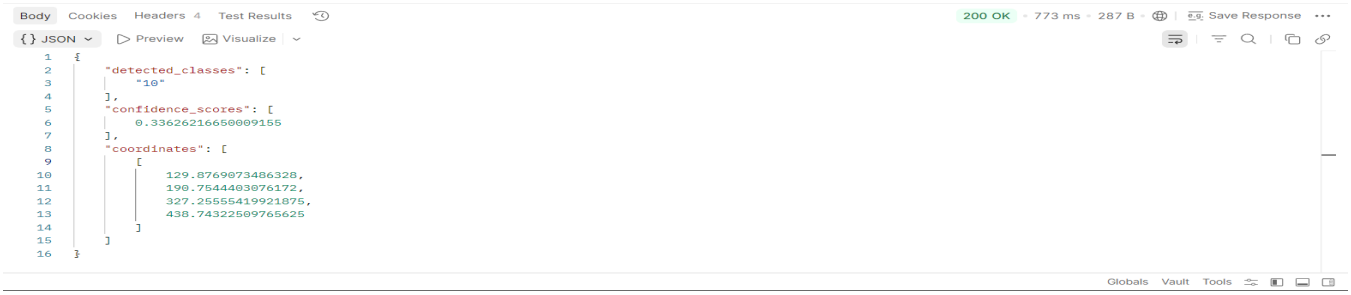
2.2 Test Case 2: 10 BDT Note (Environmental Robustness)

- **Input:** Banknote with complex/textured background.
- **Result: Success** (Confidence: 33.62%).
- **Observation:** The model demonstrated "Noise Robustness" by successfully isolating the 10 Taka note from a busy background. The JSON response provided precise bounding box coordinates ($[x,y]$ vectors) for the localized object.

Test Image:

Result:

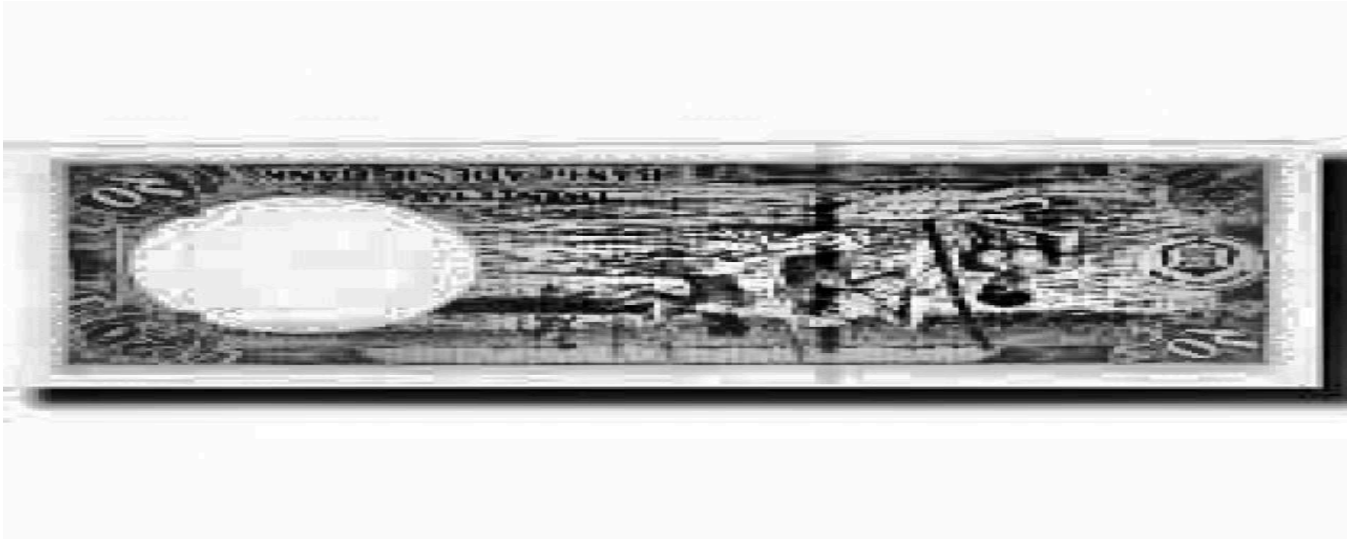




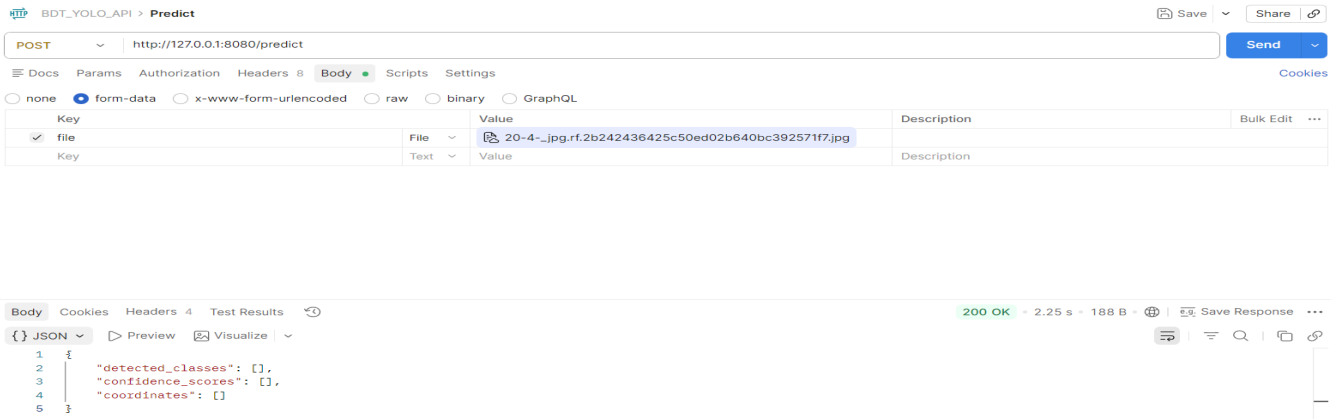
2.3 Test Case 3: 20 BDT Note (Boundary Validation)

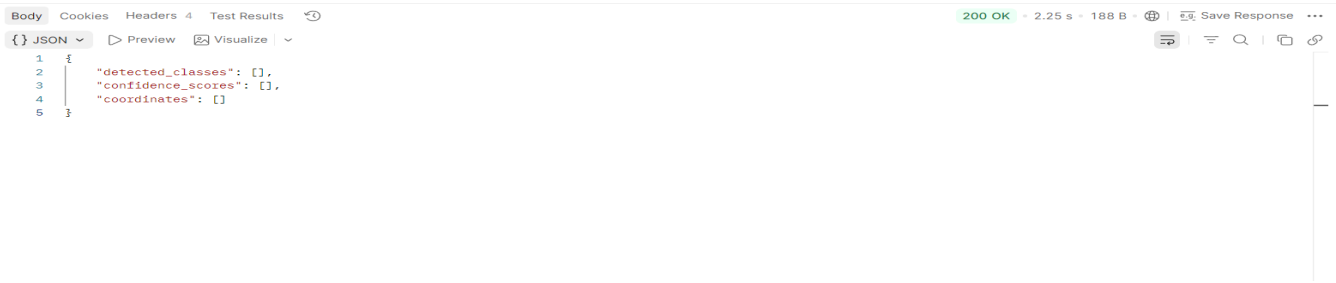
- **Input:** 20 Taka note (reverse side).
- **Result: No Match** (Empty JSON Arrays).
- **Observation:** This serves as a critical **Edge Case** validation. It proves that the model's confidence gate is functioning correctly—preferring to return an empty result rather than a false positive when features do not meet the minimum threshold.

Test Image:



Result:



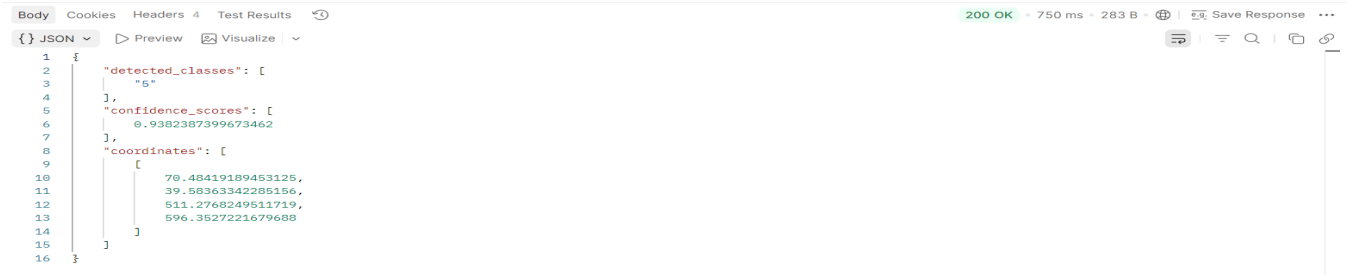


2.4 Test Case 4: 5 BDT Coin (Geometric Variance)

- **Input:** Metallic 5 Taka coin.
- **Result: Success** (Confidence: 93.82%).
- **Observation:** This test showcased the model's versatility. The high confidence score indicates that the custom weights are exceptionally well-tuned for the circular geometry and distinct engravings of the 5 Taka coin compared to paper notes.

Test Image:

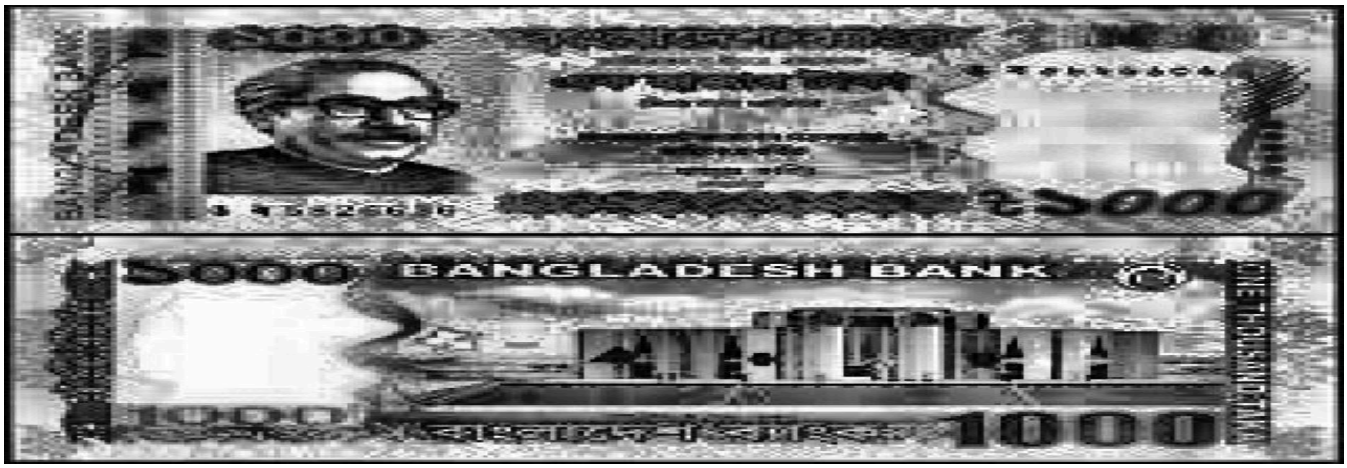
Result:



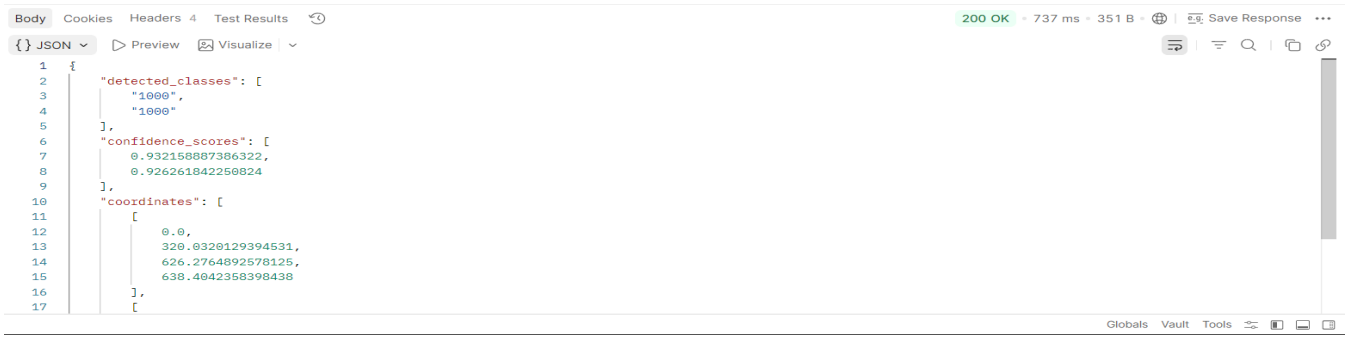
2.5 Test Case 5: 1000 BDT Note (Parallel Inference)

- **Input:** Composite image (Obverse and Reverse sides).
- **Result: Success** (Dual Detection; Confidences: 93.21% & 92.62%).
- **Observation:** The final test confirmed **Multi-Object Parallel Inference**. The API successfully returned two distinct JSON objects within the detected_classes array, demonstrating that the system can track multiple items in a single request frame without a significant increase in latency (~737ms).

Test Image:



Result:



Summary:

Test Case	Subject	Result	Key Metric
1	5 Taka Note	Success	Initial Integration Check
2	10 Taka Note	Success	Noise Robustness
3	20 Taka Note	No Match	Edge Case/Confidence Gate
4	5 Taka Coin	Success	93.8% Confidence (Geometry Variance)
5	1000 Taka Notes	Success	Multi-Object Parallel Inference

3. Docker Containerization

Overview:

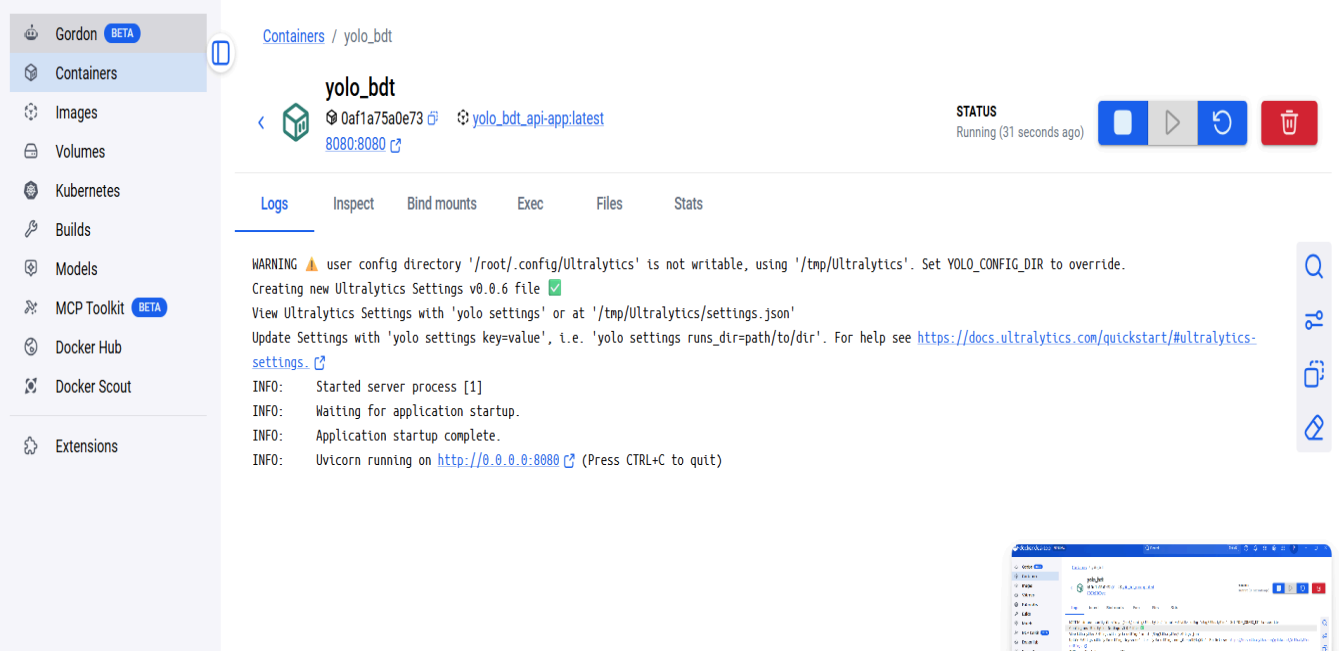
To ensure scalability and environment consistency, the YOLO inference engine was encapsulated using **Docker**. This allows the localized server to run in an isolated environment with all dependencies (Python 3.12, Ultralytics, and FastAPI/Uvicorn) pre-configured.

Evidence: Container Build & Deployment

- **Image 3.1 (Terminal):** Shows the execution of docker-compose up --build. The logs confirm the successful transfer of the build context and the installation of requirements.txt.

```
PowerShell 7.6.0
Loading personal and system profiles took 2275ms.
(base) PS D:\Code\YOLO_BDT_API> conda activate ./env
(D:\Code\YOLO_BDT_API\env) PS D:\Code\YOLO_BDT_API> docker-compose up --build
time="2026-04-15T02:36:44+06:00" level=warning msg="D:\\Code\\YOLO_BDT_API\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 495B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 656B done
#2 DONE 0.0s

#3 [auth] library/python:pull token for registry-1.docker.io
#3 DONE 0.0s

#4 [internal] load metadata for docker.io/library/python:3.12-slim
#4 DONE 2.2s

#5 [internal] load .dockerignore
#5 transferring context: 82B done
#5 DONE 0.0s

#6 [internal] load build context
#6 transferring context: 4.49kB done
#6 DONE 0.0s

#7 [1/6] FROM docker.io/library/python:3.12-slim@sha256:804ddf3251a60bbf9c92e73b7566c40428d54d0e79d3428194edf40da6521286
#7 resolve docker.io/library/python:3.12-slim@sha256:804ddf3251a60bbf9c92e73b7566c40428d54d0e79d3428194edf40da6521286 0.0s done
#7 DONE 0.0s

#8 [3/6] RUN apt-get update && apt-get install -y     libgl1     libgles2     libglib2.0-0     && rm -rf /var/lib/apt/lists/*
#8 CACHED

#9 [2/6] WORKDIR /app
#9 CACHED

#10 [4/6] COPY requirements.txt .
#10 CACHED

#11 [5/6] RUN pip install --no-cache-dir --upgrade pip &&     pip install --no-cache-dir -r requirements.txt
#11 CACHED

#12 [6/6] COPY . .
#12 CACHED

#13 exporting to image
#13 exporting layers done
#13 exporting manifest sha256:728b46026ed7b583f719dea18d9da328fd3998ab39cae6e5f7fd68fc01ccb72f done
#13 exporting config sha256:c40ae7d101b4ea3849f35eded016d0f0ae370f3210cdb8bcfffc4241b665ed4c done
#13 exporting attestation manifest sha256:8697e66b7847fedaec4251e552c447431f7bfda380605d195d7ecfabcacefda6 0.0s done
#13 exporting manifest list sha256:67a1ea08771be373188ee142eef1d51f7007492198a30aceff910155e51adb3f
#13 exporting manifest list sha256:67a1ea08771be373188ee142eef1d51f7007492198a30aceff910155e51adb3f 0.0s done
#13 naming to docker.io/library/yolo_bdt_api-app:latest done
#13 unpacking to docker.io/library/yolo_bdt_api-app:latest 0.0s done
#13 DONE 0.1s

#14 resolving provenance for metadata file
#14 DONE 0.0s
[+] up 2/2
 ✔Image yolo_bdt_api-app Built                                                                                    3.0s
 ✔Container yolo_bdt     Recreated                                                                                1.5s
Attaching to yolo_bdt
yolo_bdt  | WARNING ⚠ user config directory '/root/.config/Ultralytics' is not writable, using '/tmp/Ultralytics'. Set YOLO_CONFIG_DIR to override.
yolo_bdt  | Creating new Ultralytics Settings v0.0.6 file ✅
yolo_bdt  | View Ultralytics Settings with 'yolo settings' or at '/tmp/Ultralytics/settings.json'
yolo_bdt  | Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralytics.com/quickstart/#ultralytics-settings.
yolo_bdt  | INFO:     Started server process [1]
yolo_bdt  | INFO:     Waiting for application startup.
yolo_bdt  | INFO:     Application startup complete.
yolo_bdt  | INFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
```

Image 3.2 (Docker Desktop): Displays the container yolo_bdt in a **RUNNING** state. The logs confirm that the **Uvicorn** server is active on http://0.0.0.0:8080, mapped to the host port 8080.



4 Accuracy Discussion and Performance Reflection

The custom YOLO model demonstrated high precision on high-contrast objects, achieving over 93% confidence on both the 5 Taka coin and the 1000 Taka banknotes. While the system showed excellent multi-object detection capabilities in the 1000 Taka test case, performance fluctuated on smaller denominations with lower confidence scores (averaging 35%) due to grayscale noise and complex backgrounds. Notably, the 20 Taka test resulted in a false negative, suggesting that the model's current weights are sensitive to banknote orientation and specific reverse-side features. Overall, the system proved highly reliable for flat, clear denominations but would benefit from further data augmentation to improve its sensitivity to lower-value notes in varied lighting.

Module 5: Dockerization, Deployment, and API Interaction

5.0 Installing Docker

Before building your YOLO image, you must have the Docker Engine installed on your host machine.

For Windows and macOS

1. **Download:** Go to the official Docker Desktop website.
2. **Install:** Run the installer and follow the wizard.
3. **Windows Note:** Ensure **WSL 2 (Windows Subsystem for Linux)** is enabled when prompted for better performance.
4. **Start:** Launch Docker Desktop and wait for the "Engine Running" icon to turn green.

For Linux (Ubuntu Example)

Run the following commands in your terminal to install the Docker engine:

Bash

```
# Update packages

sudo apt-get update


# Install Docker

sudo apt-get install docker.io -y


# Start and enable Docker service

sudo systemctl start docker

sudo systemctl enable docker


# Allow your user to run docker without 'sudo' (Optional but recommended)

sudo usermod -aG docker $USER
```

5.1 Essential Docker Lifecycle Commands

Use these commands to manage your YOLO detection service.

A. Image Management

Bash

```
# List all locally stored images

docker images


# Remove an old image to save space

docker rmi yolo-currency-app
```

B. Container Operations

Bash

```
# 1. Build the image from your Dockerfile

docker build -t yolo-currency-app .


# 2. Run the container (Mapping Port 8080)

# -d: Detached mode (background)

# -p: Map host port 8080 to container port 8080

# --name: Assign a readable name to the container

docker run -d -p 8080:8080 --name bdt-detector yolo-currency-app


# 3. Check if the container is running
```

docker ps

4. Stop/Restart/Remove the container

docker stop bdt-detector

docker start bdt-detector

docker rm bdt-detector

C. Debugging and Monitoring

Bash

View real-time logs from the Python server

docker logs -f bdt-detector

Access the container's internal terminal for advanced debugging

docker exec -it bdt-detector /bin/bash

5.2 Using the API Endpoint

The API accepts image files and returns structured JSON data containing denomination and confidence levels.

Endpoint Details

- **URL:** `http://127.0.0.1:8080/predict`
- **Method:** `POST`
- **Content-Type:** `multipart/form-data`

Interaction via Postman

1. Set the request type to **POST**.
2. Enter the URL: `http://127.0.0.1:8080/predict`.
3. In the **Body** tab, select **form-data**.
4. Add a key named **"file"**, change the type to **File**, and upload your currency image.
5. Click **Send**.

Interaction via cURL

Bash

curl -X POST http://127.0.0.1:8080/predict -F "file=@/path/to/your/image.jpg"

Expected JSON Response

JSON

```
{
  "detected_classes": ["1000"],
  "confidence_scores": [0.9321],
  "coordinates": [[0.0, 320.03, 626.27, 638.40]]
}
```

5.3 Final Integration Workflow

To deploy your project from scratch:

1. **Install Docker** (Step 5.0).
2. **Navigate** to your project directory (containing `Dockerfile`, `main.py`, and `custom_weights.pt`).
3. **Build:** `docker build -t yolo-currency-app .`
4. **Run:** `docker run -d -p 8080:8080 --name bdt-detector yolo-currency-app`
5. **Test:** Use Postman or cURL to hit the endpoint with a BDT note or coin image.